

# Simba Corp Graduate Engineer Trainee - Interview & Written Prep Guide

## Schedule & Important Information

| Parameter            | Details                                          |
|----------------------|--------------------------------------------------|
| Date                 | Friday, 2nd October 2026                         |
| Written Test         | Be seated by 9:00 AM                             |
| Individual Interview | 11:30 AM – 12:00 PM (Candidate #6)               |
| Venue                | University Main Campus Engineering Boardroom     |
| Requirements         | Original stamped and signed academic transcripts |

## Part 1: Core Technical Review & High-Yield Principles

### 1. Internal Combustion Engines (ICE) & Thermodynamics

- **Four-Stroke Operating Cycles:**
  - **Otto Cycle:** Constant volume heat addition ( $PV^k = C$  isentropic compression and expansion). Thermal efficiency is given by  $\eta = 1 - \frac{1}{r^{\gamma - 1}}$ , where  $r$  is the compression ratio (typically 8:1 to 12:1).
  - **Diesel Cycle:** Constant pressure heat addition. Higher compression ratios (15:1 to 22:1) yield higher thermal efficiency due to self-ignition operation.
  - **Dual / Sabathe Cycle:** Best represents modern high-speed diesel engines with common-rail direct injection, combining constant volume and constant pressure heat addition processes.
- **Valvetrain Dynamics & Breathing:**

- **Valve Overlap:** Period during which both intake and exhaust valves remain open simultaneously around TDC to exploit exhaust gas column inertia for scavenging and exhaust cooling.
- **Volumetric Efficiency ( $\eta_v$ ):** Ratio of actual air mass inducted to theoretical swept volume displacement under ambient conditions. Maximized using turbocharging, variable valve timing (VVT), and tuned intake runners.
- **Fuel Injection & Combustion Phenomenon:**
  - **Knock / Pre-ignition (SI Engines):** Uncontrolled auto-ignition of unburned end-gas ahead of the spark flame front, caused by high compression temperatures, excessive spark advance, or low octane fuel.
  - **Diesel Knock (CI Engines):** Result of an excessively long ignition delay period leading to sudden spontaneous ignition of a large accumulated fuel mass, causing a rapid rate of pressure rise ( $dP/dt$ ).
  - **Common Rail Direct Injection (CRDI):** High-pressure accumulator rail (1,600 to 2,500+ bar) enabling precise multi-stage injection events (pilot, main, post) to minimize noise, vibration, and emissions.

## 2. Automotive Chassis, Powertrain & Dynamics

- **Transmission Systems:**
  - **Manual & Dual-Clutch (DCT):** DCT uses two concentric input shafts (odd and even gear sets) with dual wet/dry clutches to execute shifts with minimal torque interruption.
  - **Planetary Automatic (AT):** Hydrodynamic torque converter with a lock-up clutch coupled to epicyclic planetary gearsets, actuated via hydraulic clutches and brake bands.
  - **Continuously Variable Transmission (CVT):** Push-belt or steel chain operating across variable-width conical pulleys for seamless, step-less ratio variation.
- **Differential & Final Drive:**
  - **Open Differential:** Splits torque equally (50/50) between driven wheels while allowing different rotational speeds ( $N_{\text{cage}} = \frac{N_{\text{left}} + N_{\text{right}}}{2}$ ). If one wheel loses traction, total drive torque is limited by the slipping wheel.
  - **Limited-Slip Differential (LSD) / Torque Vectoring:** Uses mechanical clutch packs, Torsen gearing, or electronic braking to transfer available torque to the wheel with higher grip.
- **Steering & Wheel Alignment:**

- **Ackermann Steering:** Geometry ensuring the inner wheel turns at a sharper angle than the outer wheel during cornering to prevent tire scrubbing ( $\cot\theta_{\text{outer}} - \cot\theta_{\text{inner}} = \frac{\text{track}}{\text{wheelbase}}$ ).
- **Caster:** Backward tilt of the steering axis (positive caster) provides directional stability and self-centering return force.
- **Camber:** Inclination angle of the wheel's vertical axis relative to the ground, optimizing the cornering contact patch.
- **Toe:** Inward (toe-in) or outward (toe-out) alignment of the tires relative to the longitudinal vehicle centerline.

### 3. Electric Vehicles (EV) & Electrification

- **Traction Motors:**
  - **Permanent Magnet Synchronous Motors (PMSM):** High power density and exceptional efficiency across variable speed ranges, ideal for urban and stop-and-go driving.
  - **AC Induction Motors (IM):** Robust design, lower manufacturing cost, no rare-earth material dependency, and zero drag losses during unpowered coasting.
- **Battery Technology & BMS:**
  - **Chemistries:** Lithium Iron Phosphate (LFP) offers superior thermal stability and long cycle life; Nickel Manganese Cobalt (NMC) provides higher gravimetric and volumetric energy density.
  - **Battery Management System (BMS):** Continuously monitors State of Charge (SoC), State of Health (SoH), cell temperatures, controls cell balancing, and oversees safety interlocks such as the High-Voltage Interlock Loop (HVIL).
- **Power Electronics:**
  - **Inverter:** Converts DC power from the battery pack into three-phase variable-frequency AC using PWM and high-speed SiC or IGBT switches.
  - **DC-DC Converter:** Steps down high traction voltage (400V/800V) to standard 12V auxiliary power for low-voltage vehicle electronics.
  - **Regenerative Braking:** Reverses motor operation to act as a generator during deceleration, converting kinetic energy into electrical energy stored back in the battery.

### Mechanics of Materials:

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- **Normal Stress:**  $\sigma = P / A$
  - **Shear Stress:**  $\tau = (V \cdot Q) / (I \cdot t)$
  - **Torsion Formula:**  $T / J = \tau / r = (G \cdot \theta) / L$ . Maximum shear stress in a solid circular shaft:  $\tau_{\max} = (16 \cdot T) / (\pi \cdot d^3)$
  - **Bending Formula:**  $\sigma_b = (M \cdot y) / I$
  - **Factor of Safety (FoS):**  $\text{FoS} = \text{Yield Strength} / \text{Working Stress}$
  - **Fluid Mechanics & Heat Transfer:**
    - **Bernoulli Equation:**  $P_1 + \frac{1}{2}\rho v_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g z_2 = \text{constant}$
    - **Radiator & Cooling Systems:** Pressurized cooling loops elevate the coolant boiling point to prevent localized vapor lock, cavitation, and thermal hotspotting.
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## Part 2: Practice Technical Exam Questions & Solutions

### Question 1 (ICE & Thermodynamics)

- **Question:** In a four-stroke turbocharged diesel engine, what is the primary objective of valve overlap?
- **Options:**
  - A) To compress residual combustion gases back into the cylinder
  - B) To utilize exhaust kinetic energy to scavenge residual gases and cool the combustion chamber
  - C) To prevent the piston from reaching TDC under high load
  - D) To delay fuel atomization until exhaust stroke ends
- **Correct Answer: B** (Scavenging and thermal management of exhaust valves and combustion chamber).

### Question 2 (ICE Combustion)

- **Question:** What primarily causes "diesel knock" in a compression ignition engine?
- **Correct Answer:** An excessively long ignition delay period. This causes an excessive accumulation of unburnt fuel in the chamber, which ignites simultaneously once auto-ignition begins, leading to an extreme rate of pressure rise ( $dP/dt$ ).

### Question 3 (Engine Calculation)

- **Question:** An engine cylinder has a bore of 80 mm, a stroke of 90 mm, and a clearance volume of 50 cm<sup>3</sup>. Find the compression ratio (r).
- **Solution:**
  - Bore (d) = 8 cm, Stroke (L) = 9 cm, Clearance Volume (V<sub>c</sub>) = 50 cm<sup>3</sup>
  - Swept Volume (V<sub>s</sub>) =  $(\pi / 4) \times d^2 \times L = (\pi / 4) \times 8^2 \times 9 = 452.39 \text{ cm}^3$
  - Compression Ratio (r) =  $(V_s + V_c) / V_c = (452.39 + 50) / 50 = \mathbf{10.05:1}$

### Question 4 (Powertrain Dynamics)

- **Question:** A vehicle with an open differential has a driveshaft turning at 1,200 RPM with a final drive ratio of 4:1. In a turn, the inner wheel rotates at 260 RPM. What is the speed of the outer wheel?
- **Solution:**
  - Differential cage speed ( $N_{\text{cage}}$ ) =  $\frac{1,200}{4} = 300 \text{ RPM}$
  - Using  $N_{\text{cage}} = (N_{\text{inner}} + N_{\text{outer}}) / 2$ :
  - $300 = (260 + N_{\text{outer}}) / 2$
  - $600 = 260 + N_{\text{outer}}$
  - $N_{\text{outer}} = \mathbf{340 \text{ RPM}}$

### Question 5 (Electric Vehicles)

- **Question:** What is the primary role of the High-Voltage Interlock Loop (HVIL)?
- **Correct Answer:** HVIL is a continuous low-voltage safety circuit routed through all high-voltage connectors and access covers. If any high-voltage connection or service disconnect is opened or compromised, the HVIL loop breaks, signaling the BMS to immediately open the main traction contactors and isolate high-voltage power.

### Question 6 (EV Energy Calculation)

- **Question:** A 75 kWh EV battery pack charges from 20% to 80% State of Charge (SoC) in 20 minutes via DC fast charging. What is the average power delivered?
- **Solution:**
  - Energy transferred =  $75 \text{ kWh} \cdot (0.80 - 0.20) = 45 \text{ kWh}$
  - Time (t) =  $20 \text{ minutes} = \frac{1}{3} \text{ hour}$
  - Average Power =  $\frac{45 \text{ kWh}}{1/3 \text{ h}} = \mathbf{135 \text{ kW}}$

### Question 7 (Mechanics Calculation)

- **Question:** A solid circular steel driveshaft of diameter 40 mm transmits a torque of 500  $\text{N}\cdot\text{m}$ . Calculate the maximum shear stress.
  - **Solution:**
    - $\tau_{\text{max}} = \frac{16 \cdot T}{\pi \cdot d^3}$
    - Given:  $T = 500 \text{ N}\cdot\text{m}$ ,  $d = 0.040 \text{ m}$
    - $\tau_{\text{max}} = \frac{16 \cdot 500}{\pi \cdot (0.040)^3} = \frac{8000}{\pi \cdot 0.000064} = \frac{8000}{0.00020106} \approx \mathbf{39.79 \text{ MPa}}$
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## Part 3: Written Test & Interview Strategy

### 1. Written Test Time Management (9:00 AM)

- **Rapid First Pass:** Immediately answer all direct theoretical definitions and concepts (cycles, BMS functions, steering geometry).
- **Calculations Second Pass:** Methodically outline knowns and unknowns. Keep strict track of unit conversions (mm to m, RPM to rad/s, minutes to hours).

### 2. Boardroom Interview Tips (11:30 AM)

- **Commercial & Brand Context:** Align core technical principles with Simba Corp's primary product lines (Mitsubishi Fuso commercial vehicles, Mahindra, Proton). Be prepared to discuss commercial diesel maintenance strategies (DPF regeneration, SCR/AdBlue systems, CRDI diagnostics) and fleet electrification trends.
- **Structured Technical Communication:** Structure responses by stating the governing engineering principle first, explaining the underlying mechanism, and concluding with its practical application in vehicle design or maintenance.

## Part 4: CV-Specific Interview Questions & Model Answers

### 1. Electric Vehicle Battery Cooling Pack

- **Q:** "Walk us through your design and thermal simulation of the EV battery cooling pack. Which coolant medium did you assume, and what flow arrangement did you use?"
- **Model Answer:**
  - "In this project, I modeled an indirect liquid cooling system using serpentine cooling plates sandwiched between battery modules, utilizing a 50/50 water-ethylene glycol mixture for its balanced specific heat capacity and anti-freeze/anti-corrosion qualities. The flow arrangement was designed with parallel U-flow channels to minimize pressure drop across the pack and ensure uniform flow distribution. In SolidWorks/Inventor, I established conjugate heat transfer boundary conditions with transient heat generation based on high C-rate discharge pulses, targeting an operational cell temperature strictly between 20°C and 40°C with an inter-cell gradient under 5°C to avoid localized degradation."
- **Q:** "What is the ideal operating temperature window for Li-ion cells, and what happens mechanically and electrochemically during thermal runaway?"
- **Model Answer:**
  - "The optimal operating temperature range is between 15°C and 35°C (acceptable up to 40°C). Below 0°C, internal resistance surges and lithium plating occurs during charging, causing capacity loss and dendrite growth. Above 55°C–60°C, the Solid Electrolyte Interphase (SEI) layer begins decomposing exothermically. If heat dissipation cannot balance heat generation, self-sustaining exothermic reactions occur: the anode reacts with electrolyte, the separator melts (short-circuiting electrodes), cathode materials break down releasing oxygen, and cell venting and combustion take place—leading to cascading thermal runaway."
- **Q:** "How did you validate your thermal simulation results in SolidWorks/Inventor against real-world boundary conditions?"
- **Model Answer:**
  - "I validated the simulation setup through analytical heat transfer checks and conservative boundary definition. I calculated theoretical Reynolds and Nusselt numbers in the cooling channels using standard empirical correlations (Dittus-Bölder) to verify convective heat transfer coefficients

calculated by the solver. For boundary conditions, I applied peak ambient summer temperatures (35°C–40°C), factored in contact resistance between cell surfaces and thermal interface materials (TIM pads), and ensured grid independence by refining mesh density along thermal interfaces until temperature deltas varied by less than 2%."

## 2. Practical Manufacturing & Workshop Experience (KIRDI)

- **Q:** "You worked with laser cutting, shaft milling, and lathe operations at KIRDI. If a machined shaft has excessive runout or vibration when installed, how do you troubleshoot the machining or alignment setup?"
- **Model Answer:**
  - "I follow a systematic troubleshooting procedure:
  - 1. Workpiece Inspection: Mount the shaft on V-blocks or between centers on a lathe and use a dial test indicator (DTI) along multiple axial positions to differentiate between bent stock, eccentricity, or out-of-roundness.
  - 2. Machine & Tooling Check: Check lathe spindle runout, chuck jaw trueness, tailstock alignment, tool overhang, and cutting tool sharpness to rule out chatter or tool deflection.
  - 3. Assembly & Mounting: Verify bearing seat clearances, keyway symmetry, coupling concentricity, and dynamic balancing before installation."
- **Q:** "How did you apply Design for Manufacturability (DFM) and tolerances when fabricating sheet-metal components?"
- **Model Answer:**
  - "At KIRDI, I applied DFM by designing parts with standard sheet metal gauges to eliminate unnecessary stock costs. For laser cutting, I observed minimum bend radii relative to sheet thickness (typically  $R \geq t$ ) to prevent outer-fiber cracking. I placed cutouts and holes at least twice the sheet thickness away from bend lines to avoid deformation during press brake forming. For tolerancing, I used ISO 2768 medium tolerances for non-critical features and tighter bilateral tolerances on mating tab-and-slot joints to minimize manual rework during assembly."
- **Q:** "Tell us about the poultry de-feathering machine. What mechanical drive mechanism did you use, and how did you size the motor?"
- **Model Answer:**
  - "We utilized a V-belt drive connecting an electric motor to the central plucking drum shaft. The V-belt system was selected because it absorbs sudden shock loads, dampens vibration, and slips safely in case of

mechanical jamming. Motor sizing was determined by calculating total torque: starting inertia of the loaded rotating cylinder plus frictional resistance of rubber plucker fingers against the birds, factored by target RPM (~300–400 RPM). Incorporating a service factor of 1.5, we selected an appropriate single-phase induction motor with sufficient starting torque."

### 3. Final-Year Project: FEA & Material Failure

- **Q:** "Explain your final-year project in simple terms: What is the significance of the Nakajima test and Forming Limit Curves (FLC) in vehicle body stamping or manufacturing?"
- **Model Answer:**
  - "The Nakajima test uses hemispherical punches with varying sheet specimen widths to subject metal to different strain states—from pure uniaxial tension to plane strain and equibiaxial tension. By mapping major and minor principal strains at the onset of necking, we generate a Forming Limit Curve (FLC). In automotive body stamping (BIW), the FLC is critical because it tells manufacturing engineers exactly how far sheet panels can be pressed and drawn without localized thinning, tearing, or splitting, avoiding costly die recuts."
- **Q:** "Why did you use the GTN-Hill'48 coupled model for additively manufactured AlSi10Mg? What type of material anisotropy or void nucleation were you modeling?"
- **Model Answer:**
  - "Additively manufactured AlSi10Mg exhibits distinct directional properties (anisotropy) due to layer-by-layer laser powder bed fusion along the build direction, alongside micro-porosities from print processing. Standard isotropic yield criteria (von Mises) fail to capture this directional dependency. Hill'48 accounts for anisotropic yield behavior across orientations, while the Gurson-Tvergaard-Needleman (GTN) micromechanical model captures ductile damage evolution—specifically void nucleation, void growth, and void coalescence leading to micro-cracking and final fracture."
- **Q:** "How does simulation save cost in tooling and die design before physical sheet metal prototyping?"
- **Model Answer:**
  - "Die tooling in automotive stamping involves expensive tool steel and precision CNC machining. Finite element simulation allows us to identify

critical failure modes—such as localized wrinkling, springback, and tearing—virtually. Engineers can optimize blankholder force, draw bead geometry, lubrication conditions, and blank outline before cutting physical tool steel, eliminating physical iteration loops and drastically reducing tooling lead time and scrap rates."

#### 4. Automotive Industry & Simba Corp Fit

- **Q:** "Simba Corp deals extensively with commercial fleets (like Fuso trucks) and passenger cars. How does your mechanical engineering background translate from CAD/simulation to diagnosing drivetrain or engine faults on the floor?"
- **Model Answer:**
  - "CAD and FEA simulations provide a deep, first-principles understanding of mechanical stress, fluid dynamics, and thermodynamics. When analyzing field failures—such as turbocharger boost leakage, crankshaft bearing scoring, or premature clutch wear—I don't just read fault codes (DTCs); I understand the mechanical mechanisms causing them. That mechanical reasoning speeds up root-cause diagnosis, whether troubleshooting common-rail fuel pressure drops or driveline vibrations."
- **Q:** "What is your understanding of the difference between servicing an internal combustion commercial vehicle versus maintenance routines on an EV powertrain?"
- **Model Answer:**
  - "ICE commercial vehicles require scheduled fluid and consumable maintenance: engine oil, fuel filters, air filters, coolant flushes, valve lash adjustments, DPF regeneration, and SCR/AdBlue maintenance. EV powertrains have significantly fewer moving parts, virtually eliminating engine/transmission mechanical service. However, EV maintenance focuses heavily on high-voltage electrical safety and diagnostics: battery thermal management coolant condition, isolation resistance testing, insulation integrity, BMS cell balancing health, power electronics cooling loops, and high-voltage interlock loop checks."
- **Q:** "Where do you see yourself contributing most during your rotations in the Graduate Engineer Program?"
- **Model Answer:**
  - "I am eager to contribute in workshop diagnostics and assembly operations. My background at KIRDI gave me an appreciation for practical shop-floor operations, assembly precision, and machining tolerances. I want to build strong diagnostic proficiency on commercial

drivetrains and assembly lines first, while also offering strong value in technical fleet analysis, telematics data, and Simba Corp's future EV product deployments."

## 5. Behavioral & Leadership

- **Q:** "You coordinated workshop and exhibition planning for WIESS 4.0. Tell us about a time you handled a conflict or unexpected delay during that summit and how you resolved it."
- **Model Answer:**
  - "During WIESS 4.0, an exhibitor arrived late with a display setup that clashed with safety corridors and power feed allocations, causing friction with adjacent student teams. Instead of escalating, I held a quick on-site meeting, reviewed the floor layout plan, identified an underutilized corner space with dedicated power access, and helped their team relocate swiftly. We kept the aisleways clear, met campus safety protocols, and allowed all exhibitors to open on schedule."
- **Q:** "Why Simba Corp specifically for your Graduate Engineer Trainee program, and how does your long-term career align with automotive engineering?"
- **Model Answer:**
  - "Simba Corp stands out as an established market leader in Kenya's automotive and mobility space, representing world-class brands like Mitsubishi Fuso, Mahindra, and Proton, with robust local assembly and service infrastructure. As a mechanical engineering graduate passionate about both commercial vehicle engineering and electric mobility, Simba Corp offers the exact hands-on breadth—from heavy commercial transport to modern passenger vehicles—where I can develop into a well-rounded automotive engineer and contribute to the company's operational growth."